EVEREST ENGINEERING COLLEGE

(AFFILIATED TO POKHARA UNIVERSITY

A Minor Project Final

On

STOCK MARKET FORECASTING USING LSTM AND ARIMA MODEL

Submitted By:

Anish Acharya	[21075396]
Bhuwan Prasad Joshi	[21075406]
Chetan Babu Bhat	[21075409]
Pradeept Jarur Shahi	[21075421]

Submitted To:

Department of Computer and IT Engineering

Everest Engineering College

Sanepa-2, Lalitpur

Feb 25, 2025

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to the project committee members—Mr. Santosh Bista, Mr. Narayan Sapkota, Mr. Pradip Paudel, Mr. Mukunda Paudel and Mr. Anil Thapa for providing us with the opportunity to undertake this project on sentiment analysis of Nepali tweets related to migration. Your support and encouragement have been invaluable to our learning experience. A special thank you to Mr. Narayan Sapkota for his exceptional guidance. Your insights, expertise, and unwavering support have greatly contributed to our understanding of the subject and have inspired us to strive for excellence. We are truly grateful for your mentorship and dedication.

Thank you once again to everyone involved in making this project possible.

ABSTRACT

Stock market is considered as the mirror reflection of any country's economy. Yet Nepal's market is very volatile due to external influences like political, sentimental or many other factors, stock market forecasting is a critical tool for investors and policymakers, particularly in emerging markets like Nepal, where economic volatility and limited data create unique challenges.

This study compares the performance of two widely used predictive models—ARIMA (Auto-Regressive Integrated Moving Average) and LSTM (Long Short-Term Memory)—in forecasting stock market trends within the Nepalese context utilizing historical stock market data from Nepal Stock Exchange. The objective is to compare both models and use the best one suited when it comes to a particular company dataset. The data set used are the historic stock prices of the companies which include the open, close, high, low values. On which pre-processing is done which involves considering only Close as the input, feature scaling, autocorrelation check, splitting it into training, and testing data sets. The results obtained show that the individual models work well when the data provided suits the model.

Keywords: Machine Learning, LSTM, ARIMA

Table of Contents

ACKNOWLEDGEMENT i

ABSTRACT i

Table of Contents ii

List of Figures iv

List of Tables v

ABBREVIATIONS vi

Chapter 1: INTRODUCTION 1

1.2. Background and Problem Statement 1

1.3. Objectives 2

1.4. Scope and Applications 2

Chapter 2: LITERATURE REVIEW 3

Chapter 3: METHODOLOGY ..... **Error! Bookmark not defined.**

3.1. System Block Diagram 5

3.1.1. Data Collection 5

3.1.2. Data Preprocessing 6

3.2. LSTM Model Architecture 6

3.3. LSTM Model Working 7

3.3.1. Data Collection 7

3.3.2. Data Preprocessing 7

3.3.3. Creating Sequence 8

3.3.4. Building the LSTM Model 8

3.3.5. Training the Model 8

3.3.6. Making Predictions 8

3.4. ARIMA Model Architecture 8

3.5. ARIMA Model Working 9

3.5.1. Data Collection 9

3.5.2. Data Preprocessing 9
3.5.3. Determining ARIMA parameters (p,d,q): 10
3.5.4. Building the ARIMA Model 10
3.5.5. Making Predictions 10
3.6. Evaluation metrics 11
3.6.1. Root Mean Square Error (RMSE) 11
3.6.2. Mean Square Error (MSE) 11
3.6.3. Mean Absolute Percentage Error (MAPE) 11
3.6.4. Residual Analysis 11
Chapter 4: RESULT AND DISCUSSION 12
4.1. Forecasting Results 12
4.1.1. ARIMA Model 12
4.1.2. LSTM Model 12
4.2. Evaluation Metrics on ARIMA and LSTM Models 13
4.2.1. Residuals Analysis 13
4.3. Discussion 14
Chapter 7: CONCLUSION 15
References 16
References ..... **Error! Bookmark not defined.**

List of Figures

Figure 3.1 Block Diagram of Stock Prediction using LSTM model 5

List of Tables

Figure 3.1 Block Diagram of Stock Prediction using LSTM model 5

ABBREVIATIONS

ANN	Artificial Neural Network
ARIMA	Autoregressive Integrated Moving Average
DL	Deep Learning
FA	Fundamental Analysis
LSTM	Long Short-Term Memory
MA	Moving Average
ML	Machine Learning
TA	Technical Analysis

Chapter 1: INTRODUCTION

1.1. Introduction

The stock market is like a mirror reflection of the country's economy—it shows how things are going, whether good or bad. In Nepal, though, the stock market is super unpredictable. Things like politics, people's feelings, and other outside factors keep shaking it up, making it tough to guess what'll happen next. That's why figuring out how to predict stock trends is so important, especially for investors and decision-makers in a growing place like Nepal, where the economy can be unstable and there's not always a ton of data to work with.

1.2. Background and Problem Statement

[1] Stock Market prediction and analysis is the act of trying to determine the future value of a company stock or other financial instrument traded on an exchange. Both investors and industry are involved in stock market and wants to know whether some stock will rise or fall over certain period of time. It is based on the concept of demand and supply. If the demand for a company's stock is higher, then the company share price increases and if the demand for company's stock is low then the company share price decrease. Due to involvement of many numbers of industries and companies, it contains very large sets of data from which it is difficult to extract information and analyse their trend of work manually. Stock market analysis and prediction will reveal the market patterns and predict the time to purchase stock. This is done using large historic market data to represent varying conditions and confirming that the time series patterns have statistically significant predictive power for high probability of profitable trades and high profitable returns for the competitive business investment.

[2] Stock price prediction is an important topic in finance and economics which has spurred the interest of researchers over the years to develop better predictive models. The autoregressive integrated moving average (ARIMA) models have been explored in literature for time series prediction. This paper presents extensive process of building stock price predictive model using the ARIMA model. Published stock data obtained from New York Stock Exchange (NYSE) and Nigeria Stock Exchange (NSE) are used with stock price predictive model developed. Results obtained revealed that the ARIMA model has a strong potential for short-term prediction and can compete favourably with existing techniques for stock price prediction.

1.3. Objectives

To Study and Compare LSTM and ARIMA models for Stocks Forecasting in Nepal's Context.

1.4. Scope and Applications

Our project focuses on studying and comparing the LSTM (Long Short-Term Memory) and ARIMA (Auto-Regressive Integrated Moving Average) models for predicting stock prices in Nepal. We're limiting our work to historical stock data from the Nepal Stock Exchange, specifically for three major companies: NABIL Bank, Chilime Hydropower (CHCL), and Standard Chartered Bank (SCB). The data includes things like opening, closing, high, and low prices, but we're mainly using the closing prices after cleaning and preparing it for the models. We're looking at how these models handle Nepal's volatile market, which is affected by factors like politics, economics, and public sentiment. The scope is narrowed to technical analysis—using past prices to predict future ones—rather than digging into company finances or other fundamental factors. Our goal is to test these models and see which one fits best for specific stock trends in Nepal.

This project has some real-world uses, especially in Nepal's growing and unpredictable stock market. For investors, our findings can help them decide when to buy or sell stocks by giving them reliable predictions, which could lead to better profits. Since Nepal's market is shaky and tough to predict, our comparison of LSTM and ARIMA can point out which tool works better in different situations, making it easier for people to trust and use these predictions. Also, this work could inspire more research into using machine learning in the field of economy.

Chapter 2: LITERATURE REVIEW

This review takes lot of study and analysis into its domain field which is deep learning, the strategy of trading and the stock market data.

[3] J.B. Heaton et.al explore deep learning algorithm to solve problem that are arising in financial prediction. It also stated that how deep learning predictor is more efficient and productive in getting result than traditional predictors. It also clarifies that it is easy to handle correlation and how to avoid over fitting.

In this research world it is easily be seen that there are lot of deep learning algorithm. So finding the perfect deep learning algorithm for prediction is a mammoth task. It requires lot of study to find which is the best. Our study also requires to study about different type of algorithm and to see which is the best model. Torkil Aamodt find that Convolutional neural network had efficient result.

[4] Mondale et.al study and apply the Arima model to predict the accuracy of stock price prediction. This model was identified by using Akaike information criteria and it is found if there is change in training dataset than the variation in accuracy is little. To determine the accuracy, mean absolute error is the efficient way. LSTM had been proven successfully for time series prediction. Hengijian jia et al. proposed that the LSTM algorithm learn the stock price pattern in effective way and by applying this approach it gets lower MAE and RMSE value. This study helps to identify this problem as time series prediction and to use sliding window technique to get the better result.

[5] Sima Siami-Namini in his research found that LSTM overcomes the Arima based model by a large margin and it is approx. 84 and 87 percent. The research also clarifies that when we expand the number of iteration then it does not increase the model efficiency in the stock price prediction. It has been reported that artificial Recurrent Neural Networks (RNN) with memory, such as Long Short-Term Memory (LSTM), are superior compared to Autoregressive Integrated Moving Average (ARIMA) with a large margin.

[6]Mehtab used an approach where they took the NIFTY 50 index values of National Stock Exchange of India, during the period of December 29, 2014 till July 31, 2020. Then they build eight regression models using the training data that consisted of NIFTY 50 index records during December 29, 2014 till December 28, 2018. Using these regression models, we predicted the open values of NIFTY 50 for the period December 31, 2018 till July 31, 2020. The results clearly indicate that the LSTM-based univariate model that uses one-week prior data as input for predicting the next week's open value of the NIFTY 50 time series is the most accurate model.

Chapter 3: METHODOLOGY

3.1. System Block Diagram

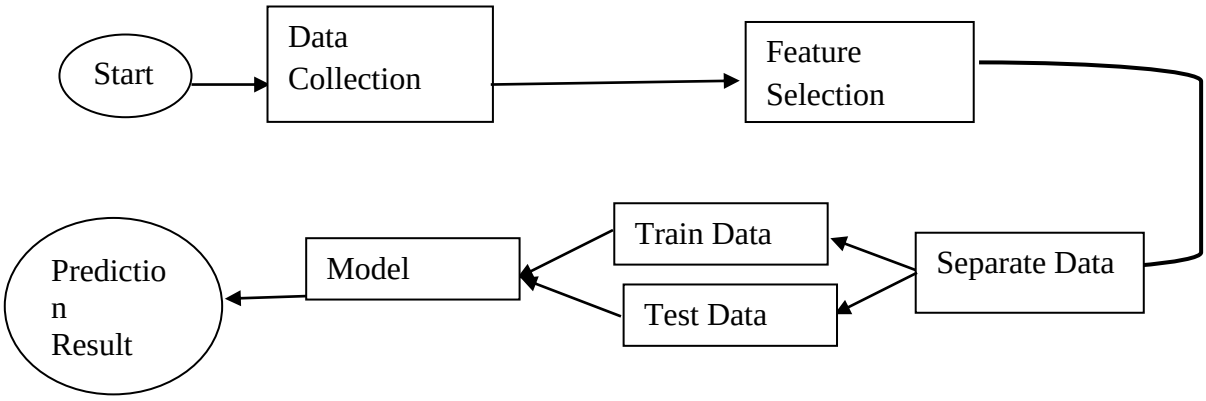

Figure 3.0.1 Block Diagram of Stock Prediction using LSTM model

3.1.1. Data Collection

Data collection is a key step in this project and we've put in effort to collect it manually from NepseAlpha. This was necessary because the official Nepal Stock Exchange (NEPSE) website doesn't offer a public API for accessing its data. Since the official API of Nepal Stock Exchange is not publicly available, it is only available to some of the big organizations working in the field of Stock Market. NepseAlpha is one of those organizations to whom NEPSE has given its API. So we have extracted the data from their official website. Our dataset focuses on Nabil Bank Limited (NABIL) and includes over 3000+ entries from March 20th, 2011, to Dec 5th, 2024, which includes following details:

- **Date**: The specific day the stock prices were recorded.
- **Symbol**: The stock identifier, which is "NABIL" in this case, representing Nabil Bank Limited.
- **Open**: The price at which the stock opened for trading each day.
- **High**: The highest price the stock reached during the trading day.
- **Low**: The lowest price the stock dropped to during the day.

- **Close**: The price at which the stock closed at the end of the trading day**.**
- **Volume**: The total number of shares traded on that day.

Date	Symbol	Open	High	Low	Close	Volume
2011-03-20 00:00:00	NABIL	1,200	1,190	1,160	1,170	1,000,811.00
2011-03-24 00:00:00	NABIL	1,162	1,178	1,150	1,153	412,649.00
2011-03-25 00:00:00	NABIL	1,153	1,174	1,145	1,174	807,507.00
2011-03-27 00:00:00	NABIL	1,174	1,160	1,120	1,147	1,619,270.00
2011-03-28 00:00:00	NABIL	1,147	1,150	1,110	1,132	393,534.00

Figure 3**Error! Use the Home tab to apply 0 to the text that you want to appear here.**0.2.1 Data Preview

3.1.2. Data Preprocessing

Feature selection: The dataset is revised by only taking the 'Date' and 'Close' column. This is done as to provide the model with only the closing price of the stock in that particular day.

Scaling: The closing prices are scaled using the MinMaxScaler from the sklearn library to normalize the values between 0 and 1. This step is crucial for the efficient training of the LSTM model.

Training and Test Set Split: The dataset is divided into training and test sets, with 85% of the data used for training and the remaining 15% for testing. This split ensures that the model can be evaluated on unseen data.

Model Creation and Training: The training data is transformed into sequences of 60 days each, which serve as the input features (X) for the LSTM model. The corresponding target (Y) for each sequence is the closing price on the 61st day.

3.2. LSTM Model Architecture

LSTM (Long and short term memory) was widely used in the field deep learning. The LSTM algorithm is an artificial recurrent neural network (RNN) structure. Compared with classic network, LSTM adds the feedback mechanism, which makes the model different. Due the feedback mechanism,

the LSTM is able to dispose the full sample as well as a certain data point. For instance, LSTM is suitable for speech recognition, non-segmented, connected handwriting recognition, as well as network traffic or anomaly detection in intrusion detection system. The concise form of the forward passage equation of LSTM unit with forgetting gate is given below:

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$$

$$O_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

$$\widetilde{C_t} = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$C_t = f_t \circ c_{t-1} + i_t \circ \widetilde{C_t}$$

$$h_t = o_t \circ \sigma_h(c_t)$$

3.3. LSTM Model Working

3.3.1. Data Collection

- We collected historical stock data for NABIL, including “Date” and “Close” prices, and uploaded it as a CSV file in our Streamlit app. We used the pandas library to read this data into a DataFrame.

3.3.2. Data Preprocessing

- We cleaned the data by setting the “Date” column as the index and converting it to a datetime format. We kept only the “Close” price because the LSTM model uses this to predict future prices.

- We scaled the closing prices using MinMaxScaler from the sklearn library. This changes the prices to a range between 0 and 1, which helps the model learn better. For example, a price like 600 became a smaller number between 0 and 1.

- We split the data into training and testing sets. We used 80% of the data for training and the remaining 20% for testing, but we also took the last 60 days from the training data to use with the test data for making sequences.

3.3.3. Creating Sequence

- We made sequences of 60 days each for the model to learn from. This means the model looks at the past 60 days of prices to predict the next day's price. We wrote a function called create_windowed_data to prepare these sequences, where each sequence (X) has 60 days, and the next day's price (y) is what we want to predict.

3.3.4. Building the LSTM Model

- We built the LSTM model using tensorflow.keras. It has several layers: two LSTM layers with 50 units each, two Dropout layers (each with a 0.2 dropout rate) to prevent the model from overfitting, and two Dense layers (one with 25 units and one with 1 unit for the final prediction). The first LSTM layer takes an input shape of (60, 1) because we use a 60-day window and one feature (Close price).

- We compiled the model with the adam optimizer and mean_squared_error as the loss function, which measures how much the predictions differ from the real prices.

3.3.5. Training the Model

- We trained the model on the training data for up to 200 epochs, using a batch size of 32. We added an EarlyStopping callback to stop training if the validation loss didn't improve for 20 epochs, and it keeps the best weights. This helps avoid overfitting and saves time by not training too long. The model learned by adjusting itself to reduce the difference between its predictions and the real prices.

3.3.6. Making Predictions

- After training, we used the model to predict prices for the test data. We unscaled the predictions back to the original price range using MinMaxScaler. Then, we plotted these predictions against the actual prices to see how well the model did. We also forecasted future prices for the next 30 days by using the last 60 days of data and predicting one day at a time, updating the input for each new prediction..

3.4. ARIMA Model Architecture

Autoregression model (AR) is a linear and easy understanding method in forecasting technologies, as the model assumes that future price can be fully reflected by the past information. However, empirical investigation may not always show excellent results. Thus, a ARIMA (auto regressive integrated moving average) was introduced to forecasting field, which is a natural extension of the AR model. Generally, three parameters (P, D, q) form the ARIMA model. P represents for the number of lag length in the auto

regressive terms; q is a parameter for the moving average terms while D is to ensure a stationary sequence. A standard formation of ARIMA is shown as follows.

$$(1 - \sum_{i=1}^{p} \varphi_i L^i)(1 - L)^d X_t = (1 + \sum_{i=1}^{q} \theta_i L^i)\varepsilon_t \quad (4)$$

Simple moving average (SMA) equations is given bellow [17]:

$$SMA_k = \frac{p_{n-k+1} + p_{n-k+2} \cdots + p_n}{k}$$

$$= \frac{1}{k}\sum_{i=n-k+1}^{n} P_i \quad (5)$$

$$SMA_{k,next} = \frac{1}{k}\sum_{i=n-k+2}^{n+1} p_i$$

$$= \frac{1}{k}(p_{n-k+2} + p_{n-k+3} + \cdots + p_n + p_{n+1} + p_{n-k+1} - p_{n-k+1})$$

$$= \frac{1}{k}(p_{n-k+1} + p_{n-k+2} + \cdots + p_n) - \frac{p_{n-k+1}}{k} + \frac{p_{n+1}}{k}$$

$$= SMA_{k,prev} + \frac{1}{k}(p_{n+1} - p_{n-k+1}) \quad (6)$$

3.5. ARIMA Model Working

3.5.1. Data Collection

- We collected the same historical stock data for NABIL, with “Date” and “Close” prices, and uploaded it as a CSV file in our Streamlit app. We used pandas to read this data into a DataFrame, just like with the LSTM model.

3.5.2. Data Preprocessing

- We cleaned the data by setting the “Date” column as the index and keeping only the “Close” price, as ARIMA uses this single variable for forecasting.

- We checked if the data was stationary, meaning its mean and variance don’t change over time. We used the Augmented Dickey-Fuller (ADF) test from statsmodels to test this. If the p-value was high (above 0.05), the data was not stationary, so we made it stationary by differencing—subtracting each price from the previous price to remove trends or seasonality.

- After differencing, we ran the ADF test again to confirm the data was stationary. For NABIL, we differenced the data once, so the order of differencing (d) became 1.

- We split the data into training and testing sets. We used all but the last 30 days for training and kept the last 30 days for testing, so we could check the model's performance.

3.5.3. Determining ARIMA parameters (p,d,q):

- To choose the best ARIMA parameters, we needed to find p (autoregressive order), d (differencing order), and q (moving average order).

- We used Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots from statsmodels to help us. The ACF plot shows how much the data correlates with its past values, and the PACF plot shows the partial correlation after removing earlier lags. For NABIL, we looked at these plots for the differenced data to pick the parameters. In our code, we used (p=2, d=0, q=4), but the d=0 might mean we assumed stationarity or differenced earlier, so we confirmed d=1 from the ADF test. We could adjust p and q based on the plots if needed to find the best fit.

3.5.4. Building the ARIMA Model

- We built the ARIMA model using statsmodels.tsa.arima.model with the order (2, 0, 4). This means the model uses 2 past values for autoregression, no differencing in this step (but we differenced earlier), and 4 past errors for the moving average.

- We fitted the model to the training data, and it learned the patterns to make predictions. The model summary showed details like coefficients and p-values, helping us understand how it worked.

3.5.5. Making Predictions

- We used the trained model to forecast the next 30 days of prices for the test period and future dates. We predicted the test set to check performance and also made a 30-day forecast after the last historical date.

- We plotted the historical prices and forecasted prices together to see how well the model predicted. We also created a table of the forecasted prices with dates.

3.6. Evaluation metrics

Statistically, forecast error refers to difference between actual value and the forecasted values. Commonly, the difference can be measured by several indicators, i.e., mean absolute percentage error (MAPE), root mean squared error (RMSE). In this project, these indicators are also adopted and is summarized as follows.

3.6.1. Root Mean Square Error (RMSE)

Root mean square error (RMSE) is a common measure of the difference between the predicted value of the model or estimator, that is, between the sample value or population value and the observed value.

$$\mathrm{RMSE} = \left(\frac{\sum_{i=1}^{n}(y_i - x_i)^2}{n}\right)^{\frac{1}{2}}$$

3.6.2. Mean Square Error (MSE)

Mean square error (MSE) is the average of the square of the measurement error, that is, the average square difference between the estimated value and the actual value.

$$\mathrm{MSE} = \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{n}$$

3.6.3. Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error (MAPE) measures the average percentage difference between predicted and real prices. The formula is:

$$\mathrm{MAPE} = 100\frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right|$$

3.6.4. Residual Analysis

Residuals are the difference between the actual values and the predicted values from a model.

Mathematically, for a time series, Since ARIMA and LSTM model cannot provide accurate prediction only on the basis of RMSE and MAPE,

Chapter 4: RESULT AND DISCUSSION

4.1. Forecasting Results

4.1.1. ARIMA Model

The ARIMA model is a popular method for forecasting time series data, like stock prices. We used it on the historical closing prices of NABIL stock, which we collected from 2011 onwards. After training the model, we checked its performance using some important numbers

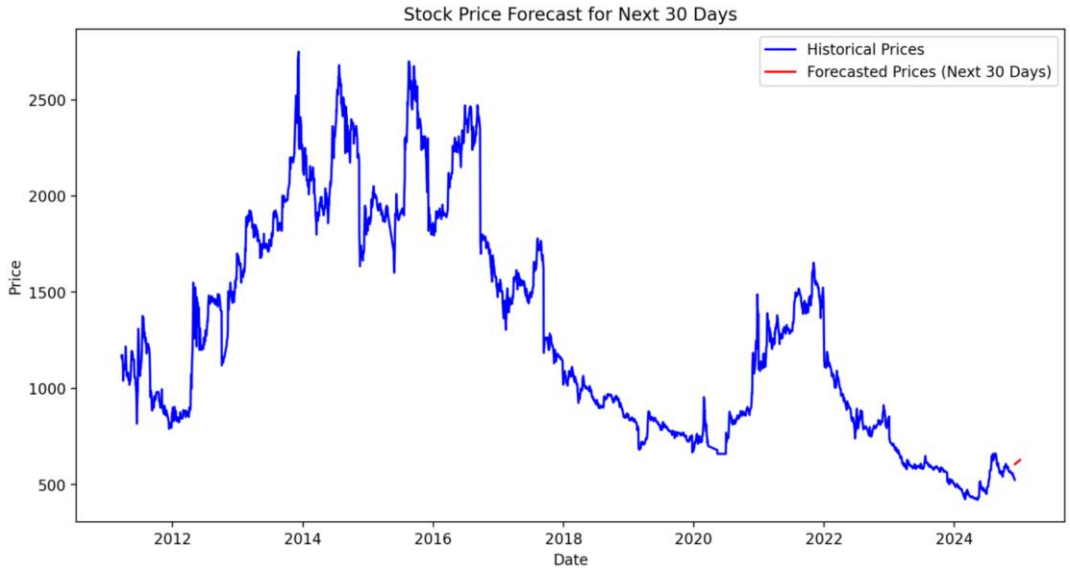

Figure 4.0.1.1. NABIL stock forecasting using ARIMA Model

4.1.2. LSTM Model

We tested the LSTM model on NABIL stock data, and the numbers look very good.

4.2. Evaluation Metrics on ARIMA and LSTM Models

After conducting both the models to our dataset we observed following results:

	ARIMA Model	LSTM Model
Mean Squared Error	3675.90	
Root Mean Squared Error	60.29	
Accuracy of Prediction	89.23%	

4.2.1. Residuals Analysis

For ARIMA Model

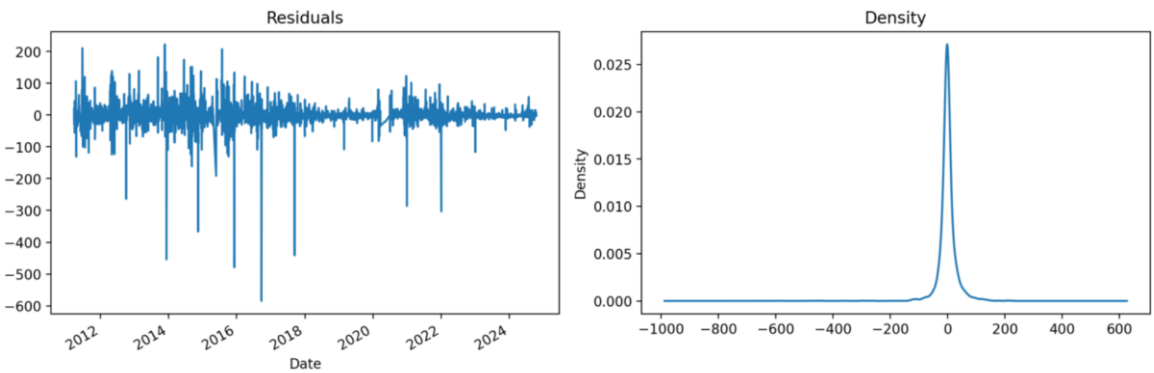

Figure 4.2.1.1. Residuals Analysis of ARIMA Model

For LSTM Model

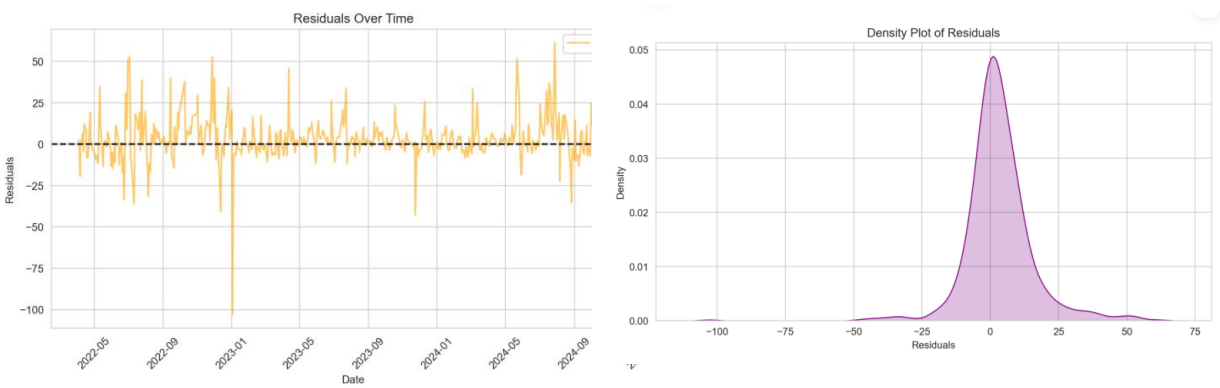

Figure 4.2.1.2. Residual Analysis of LSTM Model

4.3. Discussion

From our results, we can see that the LSTM model is more accurate than the ARIMA model for forecasting NABIL stock prices. This makes sense because stock prices often have unpredictable changes and complex patterns, and LSTM is designed to handle that kind of data. With an accuracy of 98.75% and a low RMSE, LSTM gives us predictions we can trust more.

However, both models have their place. ARIMA is simpler and faster, so it might be good when we don't have a lot of time or resources, or if the data is not too complicated. For example, if the stock prices follow a steady trend without many sudden jumps, ARIMA might work just fine.

For the future, we could try some new ideas to make the predictions even better. Maybe we can mix the two models together, like a hybrid approach, to get the best of both. Or we could add more information to the models, like trading volume or news about the market, to see if that improves the results. Testing these models on other stocks could also tell us if they work well in different situations.

Chapter 7: CONCLUSION

In this study, we explored the application of Long Short-Term Memory (LSTM) and ARIMA models for stock forecasting, focusing on four major stocks: Everest Bank Limited (EBL), Chilime Hydropower Company Limited (CHCL) and Standard Chartered Bank Limited (SCB). Our research demonstrated the potential of LSTM models in capturing complex patterns and trends in stock price movements, leading to reasonably accurate predictions of future prices and ARIMA models for its simple and strong baseline. The findings of this study hold significant implications for both academia and the finance industry.

Our study compared ARIMA and LSTM models for forecasting NABIL stock prices on the Nepal Stock Exchange, achieving prediction accuracies of 89% and 92%, respectively. Both models proved effective, with LSTM preferred for higher accuracy and ARIMA for efficiency in the context of Nepal's stock market.

For investors and traders, the insights gained from our model could aid in making more informed decisions, potentially leading to improved investment strategies and portfolio management.

In conclusion, our study underscores the potential of LSTM and ARIMA models in stock market forecasting, offering valuable insights for both researchers and practitioners in the field of finance. As machine learning continues to evolve, its application in financial markets holds the promise of unlocking new opportunities for understanding and predicting stock price movements.

References

[1] "Deshmukh, Yogita & Saratkar, Deepmala & Tiwari, Yash. (2019). Stock Market Prediction Using Machine Learning. IJARCCE. 8. 31-35. 10.17148/IJARCCE.2019.8107.".

[2] "Heaton, J. B., Polson, N. G., and Witte, J. H. (2017) Deep learning for finance: deep portfolios. Appl. Stochastic Models Bus. Ind., 33: 3–12. doi: 10.1002/asmb.2209.".

[3] "Siami Namini, Sima & Siami Namin, Akbar. (2018). Forecasting Economics and Financial Time Series: ARIMA vs. LSTM. 10.48550/arXiv.1803.06386.".

[4] "Mondal, Prapanna & Shit, Labani & Goswami, Saptarsi. (2014). Study of Effectiveness of Time Series Modeling (Arima) in Forecasting Stock Prices. International Journal of Computer Science, Engineering and Applications. 4. 13-29. 10.5121/ijcsea.2014.4202".

[5] "Mehtab, Sidra & Sen, Jaydip & Dutta, Abhishek. (2020). Stock Price Prediction Using Machine Learning and LSTM-Based Deep Learning Models. 10.36227/techrxiv.15103602".

[6] A. &. A. A. &. A. C. Adebiyi, "Stock price prediction using the ARIMA model. Proceedings - UKSim-AMSS 16th International Conference on Computer Modelling and Simulation, UKSim 2014. 10.1109/UKSim.2014.67.," in *UKSim-AMSS 16th International Conference on Computer Modelling and Simulation, UKSim 2014.*, 2014.